

TorqCapture — DIY Build Guide v1

0. What we build, what we buy

Everything the [TorqCapture — Venue Scenarios & Build Plan v1](#) §4 table marks **DIY**, turned into a shopping list, a cut list and a step-by-step. TorqCapture fabricates almost nothing physical — no gauntlet kit, no castle stock, no boarded pitch — so this guide is short: a zone kit that's mostly electronics, a soft rail, two small driver stands, and the standard pit + AV. Prices are Sept-2026 India ballparks — check the live listing before ordering. Amazon links are **search links**, not fixed listings.

Item	Scenario	Est ₹	Section
Capture zones — 3 rings, tape + optional foam lip + optional LED	All	500–15,600 (flat-taped → full lit)	§2
Scoring rig — overhead cam, markers, cabling, ± ESP32/PSU	All	6,550–27,100 (CV only → CV + LED block)	§3
Soft perimeter barrier — noodle-on-sandbag rail	A (full), B (partial)	A: 6,600–13,400 · B: 2,400–4,800	§4
Driver stands — Red + Blue	All	5,100–9,800	§5
Pit + charging table	All	12,900–24,300	§6
Scoreboard / AV	All	0–2,000	§7
Tools, if we own none	—	6,000–10,000	§1

Not in this guide (vendor): branding print, DJ/sound, food, inflatable air-wall (Scenario C), party tent (Scenario C), portable AC (Scenario C) — see the Venue Scenarios doc §4. **Not re-derived here:** the electronics BOM and the bench-test protocol — those live in [TorqCapture — Zone-Capture Tech](#) and [TorqCapture — What The Event Needs](#) §2; this guide only tells you how to physically build and rig what those docs already costed.

1. Tools + consumables — one shopping run

Buy once, reuse every event. Skip anything already in the garage.

Tools

Tool	Why	Est ₹	Where
Cordless drill/driver, 12–18 V, with bit set	Every screw in this guide	2,500–4,500	amazon.in
Hand saw (PVC + wood)	Cuts the rail stubs, rail pipe and stand-top offcuts	300–500	amazon.in

Tool	Why	Est ₹	Where
Measuring tape 5 m + carpenter's square	Layout, ring radius marking	300–600	amazon.in
F-clamps, 4 × 300 mm	Hold the stand-top ply while screwing	800–1,400	amazon.in
Utility knife + spare blades	EVA foam ring-lip cuts, cable tidy	150–300	amazon.in
Sandpaper 80 + 120 grit	Every cut ply edge gets sanded	200–400	amazon.in
Hot-glue gun + sticks	Tacking LED strip and marker-top card	300–600	amazon.in
Safety glasses + dust masks	Ply dust, glue fumes	300–500	amazon.in
Wire strippers + basic multimeter (only if wiring the ESP32/PSU box)	LED-driver build, §3	400–800	amazon.in

Consumables (shared across every build below)

Item	Qty	Est ₹	Where
Wood screws 25 mm + 40 mm, boxes	1 box each	400–700	amazon.in
Fevicol SR 998 contact adhesive (EVA to tape/turf)	1 L	400–600	amazon.in
Fevicol MR / wood glue	500 g	200–300	amazon.in
Cable ties 200 mm + 300 mm, black	100 each	300–500	amazon.in
Gaffer tape, black + coloured, 4 rolls	4	900–1,600	amazon.in
Heavy-duty adhesive velcro, 25 mm	5 m	400–800	amazon.in
Empty sandbags (fill locally)	50	1,800–3,200	amazon.in
Sand	3 bags	300–600	Local hardware / construction supplier

Local, not Amazon: 18 mm plywood (timber yard, take §8's cut plan), 25 mm PVC pipe in 3 m lengths, elbows/tees. Every Mumbai timber yard cuts to size for ₹20–40 a cut.

2. Capture zones — the ring build

Capture zone — ring build

Gaffer-taped ring ± raised foam lip ± WS2812B LED · strap-on marker top per car · ring size frozen by the bench test, not guessed here

A · RING EDGE — side section
tape only → + foam lip → + LED, in cost order

B · PLAN VIEW

pool-noodle lip (option 2, bench-test decides if it helps)
WS2812B strip + diffuser (option 3, colour flips on capture)
gaffer tape ring (option 1 — flat, ~₹0/ring)
turf / floor
cam reads this circle in software — flat tape needs no physical lip at all

Ring size (1.5–2 m) + hold-timer (~5 s) are FROZEN by the Zone-Capture Tech bench test — build to those tuned numbers. Don't cut tape or order LED before that test PASSES.

C · TEAM MARKER TOP — per fleet car (12–16 + ~4 spares)
1.5–2 m diameter
ArUco / coloured disc — velcro or zip-tie, faces up
rented car — NOTHING bolted, glued or screwed (same rule as the foam bump-rings — Car Fleet doc)
team marker top (printed)

gaffer tape / LED (owned, reused) optional foam lip

Ring size (1.5–2 m) and the ~5 s hold-timer are FROZEN by the [TorqCapture — Zone-Capture Tech bench test](#) — build to those tuned numbers, don't guess them here, and don't cut tape or order LED before that test PASSES. Three build options, cheapest first, all valid:

1. **Flat gaffer-taped ring** — the cam defines the boundary in software; ~₹0 of ring hardware.
2. **+ raised foam lip** — a low pool-noodle rim so a shoved car rebounds back in; bench-test decides if it helps or traps cars.
3. **+ WS2812B LED under a diffuser** — the content-hero upgrade, flips colour on capture.

Materials (per 3-ring Domination set — reused across every single-zone mode too)

Item	Qty	Est ₹	Where
Coloured gaffer tape — rings + Hotzone pre-taped spots	2–4 rolls	500–1,000	amazon.in
Pool noodles — raised foam lip (optional)	~9–12 for all 3 rings	1,100–3,000	amazon.in
WS2812B 5 m addressable strip — Domination 3-ring block (optional)	3–4 strips	7,500–10,400	amazon.in

Item	Qty	Est ₹	Where
WS2812B 5 m strip — single-zone modes only (optional)	1–2 strips	2,400–5,200	amazon.in
ESP32 LED-driver dev board (only with the LED build)	1–2	250–1,200	amazon.in
5V 30–40A PSU + injection taps (only with the LED build)	1 set	1,500–4,000	amazon.in
Team marker tops — 12–16 fleet cars + ~4 spares	16–20	400–1,000 (lot)	Printed card/foam, we make
Courier neutral core orb + spare (exhibition mode)	2	600–1,200	amazon.in
Flat-taped total (no LED)		≈ 500–1,000	plus the marker tops + core, §2 continued below
Full lit + foam-lip total		≈ 13,700–20,800	strips + ESP32 + PSU + lip, on top of tape

Exact totals for the whole zone kit (including the scoring camera) are in [TorqCapture — What The Event Needs](#) §2: flat-taped ≈ ₹7,650–13,700, full lit + foam lip ≈ ₹16,900–32,300. Not recomputed here — this table is the ring-only physical build.

Cut list

- Pool noodles (if building the lip): cut to arc-length segments to follow the ring curve, ~40–50 cm chords
- WS2812B strip (if building lit rings): cut to the ring's circumference — a 1.5 m ring is $\pi \times 1.5 \approx 4.7$ m, a 2 m ring is $\pi \times 2 \approx 6.3$ m, so budget one 5 m strip per ring at the small end, two at the large end
- Marker tops: printed card/foam, ~10 × 8 cm per top, one per fleet car

Steps

1. **Confirm the frozen numbers** from the bench test — ring diameter and hold-timer — before marking anything.
2. Mark the ring centre, then use a string-and-marker compass (a peg + a length of gaffer at the radius) to draw a clean circle on the turf or floor.
3. Lay the **gaffer tape** along the marked circle, overlapping every ~30 cm so it doesn't lift.
4. **Optional foam lip**: curve pool-noodle segments along the tape line, held down with the same PVC-stub-in-sandbag anchors as §4's perimeter rail. Test whether it helps a car hold position or traps it — that's the bench test's call, not a build guess.
5. **Optional LED ring**: run the WS2812B strip inside a slim diffuser channel along the tape line, wire it to the ESP32 driver box (§3), and confirm the colour flips correctly on a mock capture before the rehearsal.
6. **Print and velcro the marker tops** — one per fleet car (12–16 + ~4 spares), strapped on so it comes off clean at handback, nothing bolted to a rented car (same rule as the foam bump-rings, [Car Fleet, Scale & Protection](#)).

7. **Test:** put 2–3 cars per team in the ring, overlapping — this is the same confinement/count check from the Zone-Capture Tech bench test, just re-run at full ring size once it's built.

3. Scoring rig — overhead cam + LED driver

Scoring rig — overhead cam + LED driver

One 1080p cam on the content truss scores every zone AND feeds the big screen · ESP32 + 5V PSU only enter with the LED build

A · RIG — side view (truss/pole over the play-area centre)

Mount height + marker size are set by the bench test's full-span coverage trial (Zone-Capture Tech §bench test) — one cam vs a 2nd stitched cam is decided there, not here.

■ camera / cabling
■ ESP32 LED driver
■ 5V PSU

B · LED DRIVER BOX — only if the lit-ring upgrade is built

Cost lives in What The Event Needs §2: CV scoring ≈₹6,550–11,500 · LED-rings block ≈₹9,250–15,600 (only if built). Not re-derived here.

The camera is the whole scorer — it covers every zone at once and doubles as the content feed.

Mount height and marker size are set by the bench test's full-span coverage trial, not chosen here — one camera vs a second stitched camera for the full 3-ring Domination span is decided at the bench test, before this rig is bought or built.

Materials

Item	Qty	Est ₹	Where
Overhead camera, 1080p USB	1 (2nd only if the full-span trial requires it)	5,150–8,500	amazon.in
USB extension cable, 5–10 m	1–2	500–900	amazon.in
Camera mount / clamp for the truss or pole	1	400–800	amazon.in
Cabling, power, misc mounts	lot	1,000–2,000	Hardware store / amazon.in
ESP32 LED-driver dev board (only with the LED build)	1–2	250–1,200	amazon.in

Item	Qty	Est ₹	Where
Project enclosure, IP-rated (only with the LED build)	1	300–600	amazon.in
5V 30–40A PSU + injection taps (only with the LED build)	1 set	1,500–4,000	amazon.in
CV scoring subtotal (no LED)		≈ 6,550–11,500	Camera + markers (§2) + cabling — matches TorqCapture — What The Event Needs §2 exactly
With the Domination LED block		≈ 15,800–27,100	Adds the strips (§2) + ESP32 + PSU — same figure as the Needs doc, not recomputed

Steps

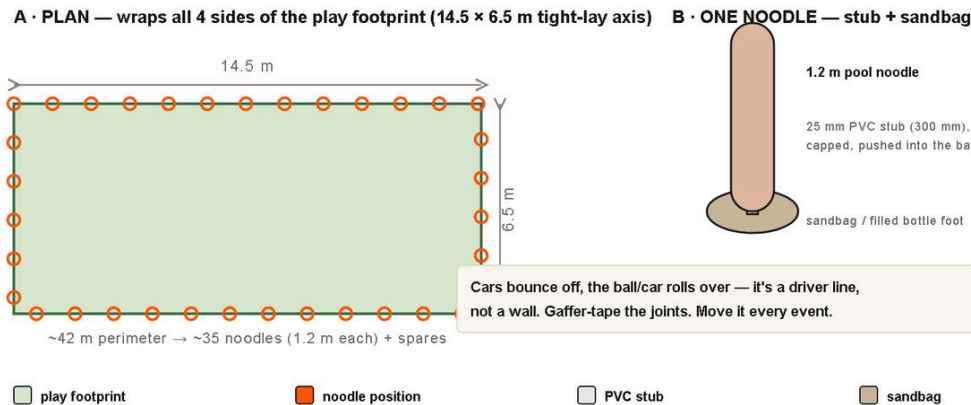
1. **Wait for the bench test PASS** before ordering the camera or any LED part — this is the one gate that changes what you buy (§0 table).
2. Rig the truss or pole over the play-area centre, at the height the bench test's full-span trial proved works for the venue's scenario.
3. Mount the camera, run the USB extension down to the show laptop, and re-run the pixel→ring calibration once in place.
4. **If building the LED rings:** wire the ESP32 inside the project enclosure — laptop feeds it the per-zone owner/state, it pushes WS2812B colour. Wire the 5V PSU with injection taps every few metres of strip run (a 14–19 m Domination run pulls real current at brightness).
5. Print + velcro the marker tops (§2) onto every fleet car in the rotation.
6. **Test:** this is exactly the Zone-Capture Tech bench-test checklist, re-run at the real venue — confinement, multi-car count, chaos/lag, cross-zone bleed, and for Domination the full-span far-ring read. A miss here means the day runs marshal-scored, not that the rig is rebuilt on the spot.

4. Soft perimeter barrier — the only enclosure

Soft perimeter barrier — the only enclosure

TORQWARS

Noodle-on-sandbag rail, FULL PERIMETER, every scenario — TorqCapture never builds hard boards (Rulebook §5)



TorqCapture never builds hard boards, in any scenario — the noodle-on-sandbag rail is the enclosure everywhere (Rulebook §5). **Scenario A needs the full loop; Scenario B only the one open side the walls/net don't already contain; Scenario C uses a rented inflatable air-wall instead (vendor, not built here).**

Materials (Scenario A — full perimeter, ~42 m around the 14.5 × 6.5 m tight-lay footprint)

Item	Qty	Est ₹	Where
Pool noodles, 1.2 m	40 (35 needed + spares)	4,800–10,000	amazon.in
25 mm PVC pipe, 3 m lengths (cut to 300 mm stubs)	4	600–1,000	Local hardware
Sandbags or 1 L filled bottles	40	1,200–2,400	—
Scenario A total (full perimeter)		≈ 6,600–13,400	

Materials (Scenario B — one open side only, ~14.5 m)

Item	Qty	Est ₹	Where
Pool noodles, 1.2 m	14 (12 needed + spares)	1,700–3,500	amazon.in
25 mm PVC pipe, 3 m lengths	2	300–500	Local hardware
Sandbags or 1 L filled bottles	14	400–850	—
Scenario B total (one side)		≈ 2,400–4,850	

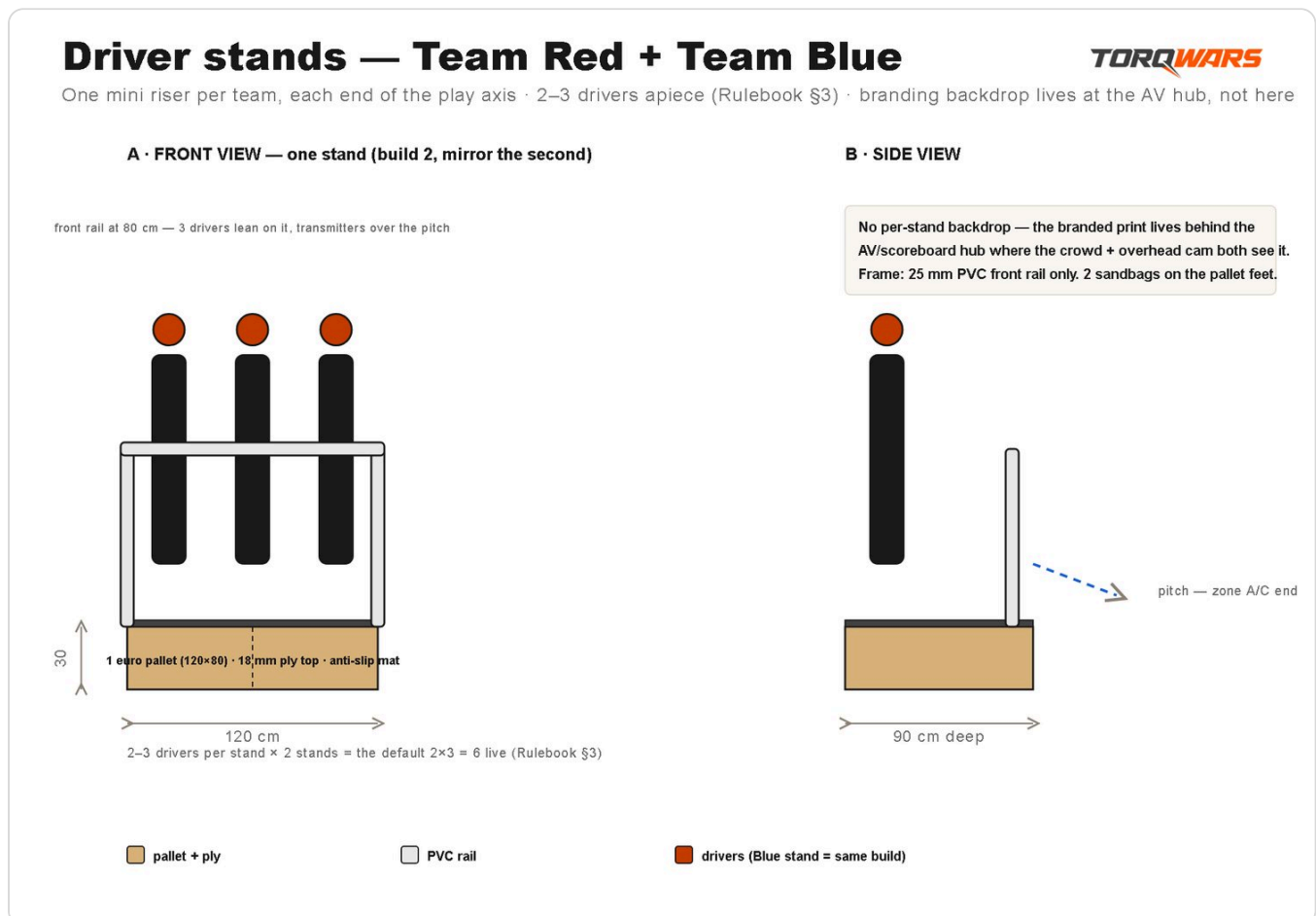
Cut list

- 25 mm PVC into 300 mm stubs, one per noodle position, capped one end

Steps

1. Cut the PVC into 300 mm stubs, cap one end.
2. Fill a sandbag to ~2 kg (or use a 1 L filled bottle), push the capped stub into the middle so it stands upright.
3. Slide a pool noodle over each stub.
4. Lay noodles end-to-end around the full perimeter of the tight-lay play footprint (Scenario A), or along the one open side only (Scenario B) — every ring stays ≥ 2.5 m off this line (Rulebook §5).
5. Gaffer-tape the joints so the line doesn't gap under a shove.
6. **Test:** push a car into the line at a few points along its length — it should bounce back, not snag or tip the noodle over.

5. Driver stands — Team Red + Team Blue



Two small risers, one per team, at opposite ends of the play axis — Red at the A end, Blue at the C end (Full Run Plan §2 09:45). Each is sized to **2–3 concurrent drivers**, matching the default 2×3 = 6 live (Rulebook §3); the branding backdrop is **not** on these stands — it lives behind the AV/scoreboard hub where the crowd and the overhead cam both see it.

Materials (for both stands)

Item	Qty	Est ₹	Where
Euro pallets, 1,200 × 800 mm, good condition	2 (1 per stand)	1,000–3,600	OLX / packaging suppliers, local
18 mm plywood, 8 × 4 ft sheet (cut per §8)	1 sheet, uncut	2,200–3,000	Timber yard
Rubber anti-slip mat, cut to 2 × 1.2 × 0.8 m portions	1 roll	800–1,400	amazon.in
25 mm PVC pipe, 3 m (front rail, both stands)	3 lengths	500–800	Local hardware
25 mm PVC tees + elbows	8	300–500	amazon.in
L-brackets 50 mm + screws	8	300–500	amazon.in
Sandbags (from §1 stock, 2 per stand)	4	—	—
Stands total (both)		≈ 5,100–9,800	

Stage-deck rental fallback: ~₹800–1,600/day per stand if the pallets don't turn up in time.

Cut list

- 18 mm ply sheet (2,440 × 1,220 mm) → 2 × stand tops, 1,180 × 780 mm each, plus an offcut for marker-top card blanks — see §8

Steps

1. Set one pallet per stand on the floor, check it sits level — shim a corner with a ply offcut if it rocks.
2. Screw the cut 18 mm ply top down into the pallet, 40 mm screws into every pallet top-board.
3. Lay the anti-slip mat, gaffer-tape the edges down.
4. **Front rail:** 2 uprights + 1 crossbar per stand in 25 mm PVC, teed into the ply edge. Friction-fit for transport, screw the tee in if it needs to be rigid.
5. One sandbag on each pallet foot.
6. Position the Red stand at the A-zone end and the Blue stand at the C-zone end of the play axis (Rulebook §5 zone spacing).
7. **Test:** 2–3 people stand on each riser and lean on the rail — nothing should flex, rock or walk.

6. Pit + charging table

Pit + charging table

One 6 ft folding table · charging LEFT, repair RIGHT · reused build — protocol is Event Ops Standards §5 + Car Fleet doc, not re-derived

6 ft × 2.5 ft folding table (180 × 75 cm) — plan view

CHARGING

LiPo

LiPo

LiPo

METAL TRAY · 2 cm of sand

packs charge HERE only — never on the table top

CHARGER

≥12 ports

LiPo BAGS

POWER STRIP (surge) · ONE wall socket · dedicated circuit

REPAIR + MARKERS

RUBBER MAT

car on its back · spares swap

MARKER BIN

spare tags/discs

velcro + zip ties

spare bumper rings

tools · hex 1.5/2/2.5

LiPo warden logs every pack in/out · hex driver set · voltage checker

ON THE FLOOR, at the charging end

ABC extinguisher

metal bin + lid

bucket of sand

Bare floor / cement board under the charging end — never turf, never the driver-stand ply.

Full LiPo protocol: Event Ops Standards §5 + Car Fleet, Scale & Protection §5 (not repeated here).

Mostly a **buy list**, not a build — the same standard TorqWars pit kit every game uses. The one rule that isn't negotiable: **LiPo packs charge on a metal tray of sand, on a bare floor, watched by one named person**. Full protocol is in [Event Ops Standards \(all games\)](#) §5 and [Car Fleet, Scale & Protection](#) §5 — not repeated here. Charger port count is driven by TorqCapture's own burn math: **≥12 packs on charge at peak** (Full Run Plan §3), so confirm the charger (or the fleet supplier's charger) covers that before buying.

Item	Qty	Est ₹	Where
Folding table, 6 ft	1 (or borrow)	3,000–5,000	amazon.in
LiPo-safe charging bags	4	800–1,600	amazon.in
Balance charger, multi-port, ≥12 ports (if the fleet doesn't bring one)	1	2,500–5,000	amazon.in
Steel tray, ~50 × 35 cm	1	400–800	amazon.in
ABC dry-powder extinguisher, 2 kg	1	1,200–2,000	amazon.in
Metal bin with lid	1	500–900	amazon.in

Item	Qty	Est ₹	Where
Surge-protected power strip, 6 socket, 3 m lead	1	800–1,500	amazon.in
Rubber work mat, 60 × 40 cm	1	400–800	amazon.in
RC hex driver set 1.5 / 2.0 / 2.5 + Phillips	1	500–1,200	amazon.in
LiPo voltage checker / cell meter	1	300–600	amazon.in
Cement board or steel sheet, ~1 m ² (floor under the tray end)	1	400–800	Local hardware
Marker/tag bin — spare tops, velcro, zip ties	1	500–1,000	We print/stock (from §2 leftovers)
Sand (from §1 stock), clipboard, marker	—	100–200	—
Pit total		≈ 12,900–24,300	

Setup (1 hour, on the day)

1. Table on **bare floor** — cement board under the charging end. Never on turf, never on the stand ply.
2. Charging end: tray with 2 cm of sand → packs lie on the sand while charging → charger beside it → LiPo bags for anything not charging. Extinguisher, sand bucket and metal bin on the floor at that end.
3. One power strip, one wall socket, dedicated circuit, cable gaffer-taped to the floor.
4. Repair end: rubber mat, marker/tag bin, tool roll.
5. Name the **LiPo warden** at the crew brief — logs every pack in/out with voltage, per Event Ops Standards §5.

7. Scoreboard / AV

Cheapest possible: a laptop and a TV/projector we already own — and it's the same laptop that runs the OpenCV scorer (Zone-Capture Tech, ₹0 for the software).

Item	Qty	Est ₹	Where
HDMI cable, 5 m	1	300–500	amazon.in
TV/projector (owned or borrowed) + stand/screen	1	0	—
Bluetooth speaker or the DJ's PA for stings	—	0	DJ vendor
AV total		≈ 0–2,000	

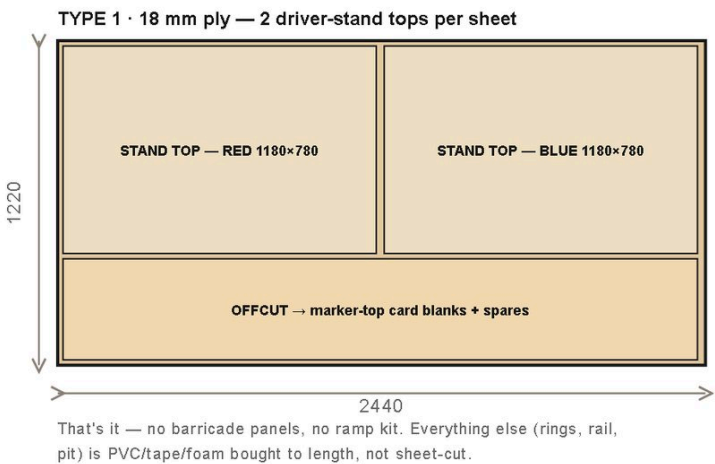
1. TV/projector beside the AV/scoreboard hub, facing the crowd, above head height.
2. Laptop on the AV table, HDMI to the screen — it's running the live scorer, not just a slideshow.
3. Test the flip-colour feed and the score tick with the DJ at sound check.

8. Plywood cut plan

Plywood cut plan



One 2440 × 1220 mm sheet covers both stands — no barricades in TorqCapture, so this is the whole cut list



That's the whole cut list — no barricade panels, no ramp kit. Everything else (rings, rail, pit) is PVC, tape or foam bought to length, not sheet-cut.

Sheet	Thickness	Yields
Type 1 × 1	18 mm	2 stand tops, 1,180 × 780 mm each (Red + Blue) + 1 offcut strip for marker-top card blanks + spares

Take this to the timber yard on D5 morning — one 8 × 4 ft sheet (2,440 × 1,220 mm), have them cut the two straight lines on the panel saw, ₹20–40 a cut.

9. Build order — D5 to D8, 2 people

Slots into the [TorqCapture — Venue Scenarios & Build Plan v1](#) \$5 timeline. Amazon orders go in the moment the bench test PASSes and the venue locks (D4), so everything lands by D6.

Day	Morning	Afternoon
D5	Timber yard with the \$8 cut plan · hardware shop for PVC, sandbags	Perimeter barrier: cut stubs, fill bags, slide noodles
D6	Driver stand pallets + ply tops screwed down	Front rails fitted · marker tops printed + velcroed
D7	Zone tape/rings marked + laid · optional foam lip fitted	Scoring rig mounted + calibrated · OpenCV bring-up (see Full Run Plan \$1 tech weekend)
D8	Dry-fit everything as one layout: rings + stands + rail + pit + AV	Pack into labelled crates · charge all batteries · print vendor confirms delivery

D10 is the full rehearsal + full-floor tech bring-up — everything built here goes up once, at full size, before the venue, per [TorqCapture — Full Run Plan](#) §1.

10. Cost roll-up

	Scenario A — raw hall	Scenario B — AC turf	Scenario C — outdoor turf
Capture zones + scoring rig (§2–§3, flat-taped → full lit)	7,650–32,300	7,650–32,300	7,650–32,300
Perimeter barrier (§4)	6,600–13,400	2,400–4,850	0 (vendor air-wall instead)
Driver stands (§5)	5,100–9,800	5,100–9,800	5,100–9,800
Pit + charging (§6)	12,900–24,300	12,900–24,300	12,900–24,300
Scoreboard / AV (§7)	0–2,000	0–2,000	0–2,000
DIY build total	≈ 32,250–81,800	≈ 28,050–73,250	≈ 25,650–68,400
Tools if none owned (§1)	+6,000–10,000	+6,000–10,000	+6,000–10,000

This is the smallest DIY build in the TorqWars games set — TorqCapture's only real new spend is the electronics (§2–§3), and even that reuses fully from event #2 onward (camera, laptop, markers, LED all strap-on or software, [TorqCapture — What The Event Needs](#)). Vendor lines (print, DJ, food, and Scenario C's tent/air-wall/AC) sit on top — see the Venue Scenarios doc §4.

11. Safety while building

- Glasses and mask for every cut and every glue-up.
- Sand every ply edge before it goes near a hand or a chassis.
- Nothing bolts, screws or glues to a rented car — marker tops are velcro/zip-tie only, exactly like the foam bump-rings ([Car Fleet, Scale & Protection](#)).
- The overhead cam mount is a rigging job: secure the clamp before running any cable, strain-relief the USB and power runs, and never route a cable across a walkway.
- LED-driver electronics (ESP32 + 5V PSU) live in an IP-rated enclosure; keep the mains feed on a dedicated, surge-protected circuit, same as the charging pit.
- LiPo protocol is not a build step — it's a person. Name the warden before anything charges (Event Ops Standards §5).